CS 171: Problem Set 3

Due Date: October 3rd, 2026 at 8:59pm

1. Two Versions of CPA security (10 points)

There are two common definitions of CPA security, which are given in definitions 0.1 and 0.2 below[1]. Prove that definitions 0.1 and 0.2 are equivalent, i.e. if a scheme is secure under one definition, then it is secure under the other definition.

Definition 0.1 *Let* $\Pi = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *be an encryption scheme and let* $\mathcal{A}$ *be an adversary for the CPA security game. Define the CPA security game as follows:*

$G_{\mathcal{A},\Pi}(n)$*:*

1. *The challenger samples a key* $k \leftarrow \mathsf{Gen}(1^n)$.
2. *The adversary* $\mathcal{A}$ *is given input* 1^n *and oracle access to* $\mathsf{Enc}(k, \cdot)$*, and outputs a pair of messages* (m_0, m_1) *with* $|m_0| = |m_1|$.
3. *The challenger samples a bit* $b \leftarrow \{0, 1\}$*, and computes the ciphertext* $c \leftarrow \mathsf{Enc}(k, m_b)$*. Then they give* c *to* $\mathcal{A}$.
4. $\mathcal{A}$ *continues to have oracle access to* $\mathsf{Enc}(k, \cdot)$ *and outputs a bit* b'.
5. *The output of the game is* 1 *if* $b' = b$*, and* 0 *otherwise.*

We say that the encryption scheme Π *is CPA-secure if for all probabilistic polynomial-time (PPT) adversaries* $\mathcal{A}$*, there is a negligible function* negl *such that*

$$\Pr\left[G_{\mathcal{A},\Pi}(n) = 1\right] \leq \frac{1}{2} + \mathsf{negl}(n)$$

In definition 0.2 below, any changes from definition 0.1 are shown in red.

Definition 0.2 *Let* $\Pi = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *be an encryption scheme and let* $\mathcal{A}$ *be an adversary for the CPA security game. Define the CPA security game as follows:*

$H_{\mathcal{A},\Pi}(n, b)$*:*

1. *The challenger samples a key* $k \leftarrow \mathsf{Gen}(1^n)$.
2. *The adversary* $\mathcal{A}$ *is given input* 1^n *and oracle access to* $\mathsf{Enc}(k, \cdot)$*, and outputs a pair of messages* (m_0, m_1) *with* $|m_0| = |m_1|$.
3. *The challenger computes the ciphertext* $c \leftarrow \mathsf{Enc}(k, m_b)$*. Then they give* c *to* $\mathcal{A}$.
4. $\mathcal{A}$ *continues to have oracle access to* $\mathsf{Enc}(k, \cdot)$ *and outputs a bit* b'.
5. *The output of the game is* b'.

[1] These are analogous to the two definitions of security for EAV security (lecture 3, slides 19-20) and PRGs (lecture 4, slides 8-9)

We say that the encryption scheme Π is CPA-secure if for all probabilistic polynomial-time (PPT) adversaries $\mathcal{A}$, there is a negligible function negl such that

$$\left| \Pr\left[H_{\mathcal{A},\Pi}(n,0)=1\right] - \Pr\left[H_{\mathcal{A},\Pi}(n,1)=1\right] \right| \leq \mathsf{negl}(n)$$

Solution

1. First, note that

$$\begin{aligned}\Pr\left[G_{\mathcal{A},\Pi}(n)=1\right] &= \Pr[b=0]\cdot\Pr\left[H_{\mathcal{A},\Pi}(n,0)=0\right] + \Pr[b=1]\cdot\Pr\left[H_{\mathcal{A},\Pi}(n,1)=1\right] \\ &= \frac{1}{2}\cdot\Big(1-\Pr\left[H_{\mathcal{A},\Pi}(n,0)=1\right]\Big) + \frac{1}{2}\cdot\Pr\left[H_{\mathcal{A},\Pi}(n,1)=1\right] \\ &= \frac{1}{2}+\frac{1}{2}\cdot\Big(\Pr\left[H_{\mathcal{A},\Pi}(n,1)=1\right]-\Pr\left[H_{\mathcal{A},\Pi}(n,0)=1\right]\Big)\end{aligned}$$

2. **Claim 0.3** *Definition 0.2 implies definition 0.1.*

   Proof If for all PPT adversaries $\mathcal{A}$, there exists a negligible function negl such that

$$\left| \Pr\left[H_{\mathcal{A},\Pi}(n,0)=1\right] - \Pr\left[H_{\mathcal{A},\Pi}(n,1)=1\right] \right| \leq \mathsf{negl}(n)$$

   then

$$\Pr\left[G_{\mathcal{A},\Pi}(n)=1\right] \leq \frac{1}{2}+\frac{\mathsf{negl}(n)}{2}$$

   Note that $\frac{\mathsf{negl}(n)}{2}$ is still a negligible function. Then for any PPT adversary $\mathcal{A}$, there exists a negligible function $\mathsf{negl}_1(n) := \frac{\mathsf{negl}(n)}{2}$ such that $\Pr\left[G_{\mathcal{A},\Pi}(n)=1\right] \leq \frac{1}{2}+\mathsf{negl}_1(n)$. ■

3. **Claim 0.4** *Definition 0.1 implies definition 0.2.*

   Proof

   (a) If for all PPT adversaries $\mathcal{A}$, there exists a negligible function $\mathsf{negl}_{\mathcal{A}}$ such that

$$\Pr\left[G_{\mathcal{A},\Pi}(n)=1\right] \leq \frac{1}{2}+\mathsf{negl}_{\mathcal{A}}(n)$$

   then

$$\Pr\left[H_{\mathcal{A},\Pi}(n,1)=1\right] - \Pr\left[H_{\mathcal{A},\Pi}(n,0)=1\right] \leq 2\cdot\mathsf{negl}_{\mathcal{A}}(n)$$

   Note that $2\cdot\mathsf{negl}_{\mathcal{A}}(n)$ is still a negligible function.

   (b) Next, we'll show that

$$\Pr\left[H_{\mathcal{A},\Pi}(n,0)=1\right] - \Pr\left[H_{\mathcal{A},\Pi}(n,1)=1\right] \leq 2\cdot\mathsf{negl}_{\mathcal{B}}(n)$$

   for some negligible function $\mathsf{negl}_{\mathcal{B}}$. Let us define a new PPT adversary $\mathcal{B}$ that runs the same algorithm as $\mathcal{A}$, except that when $\mathcal{A}$ outputs b', $\mathcal{B}$ outputs $b'\oplus 1$. Since $\mathcal{B}$ is a PPT adversary, we know that there exists a negligible function $\mathsf{negl}_{\mathcal{B}}$ such that

$$\Pr\left[H_{\mathcal{B},\Pi}(n,1)=1\right] - \Pr\left[H_{\mathcal{B},\Pi}(n,0)=1\right] \leq 2\cdot\mathsf{negl}_{\mathcal{B}}(n)$$

   Therefore

$$\begin{aligned}\Pr\left[H_{\mathcal{A},\Pi}(n,0)=1\right] - \Pr\left[H_{\mathcal{A},\Pi}(n,1)=1\right] &= \Pr\left[H_{\mathcal{B},\Pi}(n,1)=1\right] - \Pr\left[H_{\mathcal{B},\Pi}(n,0)=1\right] \\ &\leq 2\cdot\mathsf{negl}_{\mathcal{B}}(n)\end{aligned}$$

(c) Finally, let $\mathsf{negl}_1(n) = 2 \cdot \mathsf{negl}_{\mathcal{A}}(n) + 2 \cdot \mathsf{negl}_{\mathcal{B}}(n)$. Note that negl_1 is a negligible function, and $2 \cdot \mathsf{negl}_{\mathcal{A}}(n) \leq \mathsf{negl}_1(n)$, and $2 \cdot \mathsf{negl}_{\mathcal{B}}(n) \leq \mathsf{negl}_1(n)$.
We've shown that for any PPT adversary $\mathcal{A}$, there exists a negligble function negl_1 such that

$$\left| \Pr\left[H_{\mathcal{A},\Pi}(n,0) = 1\right] - \Pr\left[H_{\mathcal{A},\Pi}(n,1) = 1\right] \right| \leq \mathsf{negl}_1(n)$$

■

■

2. Feistel Network (10 points)

A Feistel network is used to construct a pseudorandom permutation F given a pseudorandom function f that is not necessarily a permutation[2]. However, if f is not pseudorandom, then F is potentially not pseudorandom either.

Consider the following three-round Feistel network given in definition 0.5 below[3].

Definition 0.5 (Three-Round Feistel Network F)

1. *Let* $f : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$.
2. ***Inputs:*** *Let* F *take as input a key* $k \in \{0,1\}^{3n}$ *and an input* $x \in \{0,1\}^{2n}$*, which are parsed as:*

$$k = (k^1, k^2, k^3) \in \{0,1\}^n \times \{0,1\}^n \times \{0,1\}^n$$
$$x = (L_0, R_0) \in \{0,1\}^n \times \{0,1\}^n$$

3. ***Computation:***
   (a) F *computes* $L_1 := R_0$ *and* $R_1 := L_0 \oplus f(k^1, R_0)$.
   (b) F *computes* $L_2 := R_1$ *and* $R_2 := L_1 \oplus f(k^2, R_1)$.
   (c) F *computes* $L_3 := R_2$ *and* $R_3 := L_2 \oplus f(k^3, R_2)$.
   (d) F *outputs* (L_3, R_3).

Suppose that there was a flaw in the design of f so that for all keys k and all inputs x, the first bit of $f(k, x)$ equals the first bit of x. Show that there exists some efficient adversary $\mathcal{A}$ that can break the pseudorandom permutation security of F *by making only a single query to* F.

Solution

1. Let a_0 be the first bit of L_0, and let b_0 be the first bit of R_0. Let $a_1, a_2, a_3, b_1, b_2, b_3$ be defined analogously. Then these bits will have the following values:

$$(a_1, b_1) = (b_0, (a_0 \oplus b_0))$$
$$(a_2, b_2) = ((a_0 \oplus b_0), a_0)$$
$$(a_3, b_3) = (a_0, b_0)$$

2. In other words, two specific output bits of F, (a_3, b_3), are equal to two specific input bits of F, (a_0, b_0). On any given input, the probability that a uniformly random permutation will produce an output pair with this property is approximately $1/4$.

3. Now we can construct our distinguisher $\mathcal{A}$ that breaks the pseudorandom permutation security of F. A will query F on an arbitrary input and check whether $(a_3, b_3) = (a_0, b_0)$. If so, $\mathcal{A}$ ouputs 1; if not, $\mathcal{A}$ outputs 0.

4. The distinguishing advantage of $\mathcal{A}$ is approximately $1 - 1/4 = .75$, which is non-negligible. Therefore, $\mathcal{A}$ breaks the PRP security of F.

■

[2]For more details, see Katz & Lindell, 3rd edition, sections 7.2.2 and 8.6.
[3]This definition is adapted from Katz & Lindell, 3rd edition, construction 8.23.